Chemistry - Graduate Level Assessment 6 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Helium effuses through a pinhole 5.33 times faster than an unknown gas. That gas is most likely
   (a) Ne
   (b) CO 2
   (c) Ar
   (d) CH 4
   (e) O_2

2. Which of the following ions possess a rare gas electronic configuration? (a) Ca^2+; (b) In^+; (c)Ga^3+; (d) Sc^3+; (e) Se^2-
   (a) In^+,Sc^3+ and Se^2-
   (b) Ca^2+, Ga^3+ and In^3+
   (c) Ca^2+, Sc^3+ and Se^2-
   (d) In^3+, Ga^3+ and Se^2-
   (e) Ca^2+, $In^+ and$Se^2-

3. Which of the following substances has an asymmetrical molecular structure?
   (a) CO
   (b) CO 2
   (c) CH 4
   (d) BF 3
   (e) PCl 5

4. Calculate theCoulombicbarrier in the bombardment of $^209 Bi Calculate the Coulombic barrier in the bom$

   5. 22.1 MeV

   6. 23.3 MeV

   7. 20.5 MeV

   8. 21.2 MeV

The atomic radius of hydrogen is 0.037 nm, Compare this figure with the length of the first Bohr radius. Explain any differences.

(a) The atomic radius of hydrogen is five times the first Bohr radius

(b) The atomic radius of hydrogen is four times the first Bohr radius

(c) The atomic radius of hydrogen is half the first Bohr radius

(d) The atomic radius of hydrogen is one tenth the first Bohr radius

(e) The atomic radius of hydrogen is larger than the first Bohr radius

True / False (10 marks)

Answer True or False

6. True or False: To decompose 454 g of water by electrolysis, 100.8 faradays of electricity is required.
7. True or False: Mass spectrometry can directly measure the smell of elements, indicating their olfactory properties.
8. True or False: (VP) / (T) = R represents the ideal gas law, illustrating the relationship between pressure, volume, and temperature of an ideal gas.
9. True or False: The ionic bond strength of NaCl is approximately 88.4% when compared to that of KF, based on ionic radii.
10. True or False: The dipole moment of formic acid (HCOOH) with an OCOH dihedral angle of $0^{\circ} iscalculatedtobe 0.85D$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the concept of half-life in the context of radioactive decay, and calculate the half-life of an unstable substance that decomposes 75% in one hour.
12. Discuss the relationship between pressure, density, and temperature in interplanetary space, and calculate the temperature required for a density of 4 molecules per cubic centimeter at a pressure of $10^{-}16torr$.

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